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Exploring the acceptance for e-learning using technology acceptance model among university students in India

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Abstract: This paper examines the acceptance and behaviour of students towards e-learning using the technology acceptance model (TAM), within the framework of a course. The instrument used in the current study adopts the scale developed by Davis (1989). This study is set against a lack of consistent, detailed research on the factors influencing the acceptance of e-learning by students in India. In line with previous research, results suggest that TAM is a strong theoretical model (Ronnie et al., 2011), where its validity can extend to the e-learning context. Significant relationships were observed between perceived ease of use, perceived usefulness, attitude, behavioural intention to use and actual use. Perceived usefulness strongly mediated perceived ease of use and attitude, while attitude mediated perceived usefulness, perceived ease of use and behavioural intention to use e-learning.

Keywords: e-learning; technology acceptance model; TAM; perceived ease of use; perceived usefulness; behavioural intention; India.

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1 Introduction

In the recent years, transmission of knowledge has been greatly influenced by information communication technology (ICT) and the internet, thereby influencing the development of e-learning. In India e-learning is widely being adopted by firms for enhancing their operations and by government for providing mass education. E-learning which generates significant performance gains for students is being introduced in the curricula of many universities in India. E-learning educates students using learning material that is fully enriched with multi-media content (Archana et al., 2013). Educators are able to communicate the teaching material to wide-spread audience from across the globe using this technology. The online self-learning support provides students with online lectures, e-text, e-exercises, e-quizzes and so on, thereby delivering higher quality in learning. However, the success of the e-learning program depends on the rate of adoption of technology by students. Understanding the student preferences, intentions and purpose for using e-learning would be help in designing and implementing better e-learning programs, thereby increasing the student acceptance for these courses.

Empirical research presents various models that investigate and understand the factors affecting the acceptance of computer technology, such as – theory of planned behaviour (TPB) (Ajzen, 1991; Mathieson, 1991); technology acceptance model (TAM) (Davis, 1989); theory of reasoned action (TRA) (Ajzen and Fishbein, 1980). TAM is an adaptation of TRA for explaining the computer usage behaviour. TAM uses TRA as a basis for specifying the causal linkages between two key beliefs – perceived usefulness and perceived ease of use; attitudes of users; their intentions; and actual computer adoption behaviour (Davis, 1989). This paper adapts TAM as it is less general and provides insights to understand the relationship between perception and usage behaviour of university students for e-learning courses. Empirical research on TAM and its application in the field of education has been reported from studies such as: examining the behavioural intention of students for using e-portfolio system (Ronnie et al., 2011); examining the acceptance for e-learning by students in Jordan (Al-Adwan et al., 2013) and so on.

1.1 Research question

TAM is a widely studied model and extensive literature is reported on TAM in the Western nations. Substantial literature exists on e-learning as well. While most consumer behaviour models are developed in the western countries, testing the applicability of these models in other cultures is essential (Durvasula et al., 1993). The principle of replicability plays a fundamental role in the research process (Hunter, 2001), as it establishes external validity of the theory. This paper attempts to explore the less researched area of the acceptance of e-learning among the university students in India. Research question for the current research has been formulated as:

- "What factors affect acceptance and behaviour of university students towards e-learning?"

The research question seeks to examine the behavioural intention of students for using e-learning. TAM aids in the above examination by seeking to understand the students' – perceived usefulness, their ease of use, their attitude towards the usage of

e-learning and influence of these factors on students' behavioural intentions to use e-learning. TAM would help in analysing the reasons for resistance or otherwise towards the technology and would further enable providers in taking efficient measures to improve user acceptance of e-learning.

The paper is organised in the following sections. A review of the literature is presented in the second section. The third section discusses the methodology adopted for conducting this research, while the fourth section discusses the analysis of data. Subsequently the findings, conclusion and directions for further research are presented in the final section.

2 Literature review

The first subsection discusses the literature on e-learning and the second discusses TAM.

2.1 E-learning

Educational institutions are leveraging technology to reach a large number of students with quality education at low cost. Many universities have integrated in their curriculum with e-learning, which is one of the many tools that have emerged from information technology. E-learning is a self-directed learning that is based on technology, especially web-based technology with an emphasis on collaborative learning (Bleimann, 2004). Definitions on e-learning refer to the use of technology as a channel for delivery of training and education. Technology here refers to the electronic media, internet, intranet, extranet, satellite broadcasts audio/video tape, inter-active TV and CD-ROM (Urdan et al., 2000), computers (Schank, 2002; Roffe, 2002; Sambrook, 2003; Tsai and Machado, 2002) and a range of other knowledge collection and distribution technologies (Fry, 2001). Advanced information technologies mediate the learner's interactions with the e-learning materials, peers and instructors (Alavi and Leidner 2001). Rosenberg (2001), states that e-learning system has the following characteristics –

1 it is based on a network

2 it focuses on the broadest view of learning

3 the participants use computers to get information and knowledge.

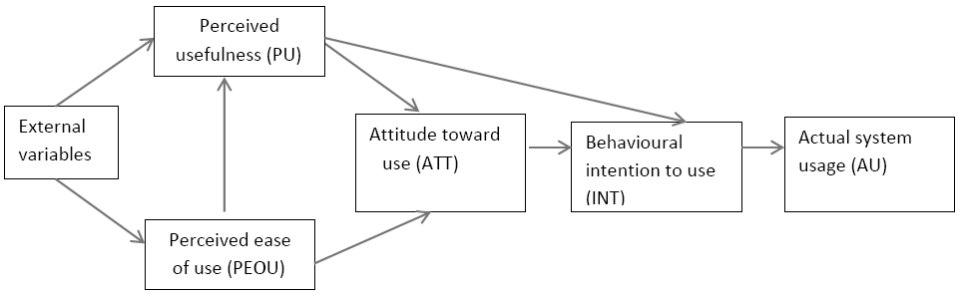
E-learning is cost-effective, flexible, has a wide reach, consistent, repeatable and convenient for users. The objectives of e-learning are dependent on the quality of the teaching process and the effectiveness of online access (Gunasekaran et al., 2002). Abrami and Barrett (2005) as referred by Ronnie et al. (2011) noted that tools such as reflective journals, self-report surveys and digital storytelling can engage learners in reflection, support learning and facilitate the creation of portfolios. Effective use of ICT in delivering e-learning based components of a course is of critical importance to the success and student acceptance of e-learning. TAM is an intention-based model developed specifically for explaining and/or predicting user acceptance of computer technology (Hu et al., 1999). In the current study, TAM is used to predict the user acceptant of e-learning.

2.2 Technology acceptance model

TAM is being adopted across different technologies, ranging from software packages to online services. Technology acceptance was defined as "an individual's psychological state with regard to his or her voluntary or intended use of a particular technology" (Hendrick and Brown, 1984). Technology acceptance may also be defined as "the demonstrable willingness within a user group to employ information technology (IT) for the tasks it was designed to support" (Dillon and Morris, 1998).

TAM was first developed by Davis (1986). TAM provides an explanation of the determinants of computer acceptance that is general and capable of explaining user behaviour across technologies and user populations (Davis et al., 1989). The model also predicts the likelihood of a new technology being adopted within a group of individuals or organisations (Davis et al., 1989). It states that the success of a system can be determined by user acceptance of the system, measured by three factors: perceived usefulness (PU), perceived ease of use (PEOU), and attitude towards usage of technology (ATT) (Figure 1).

Figure 1 Initial model of TAM



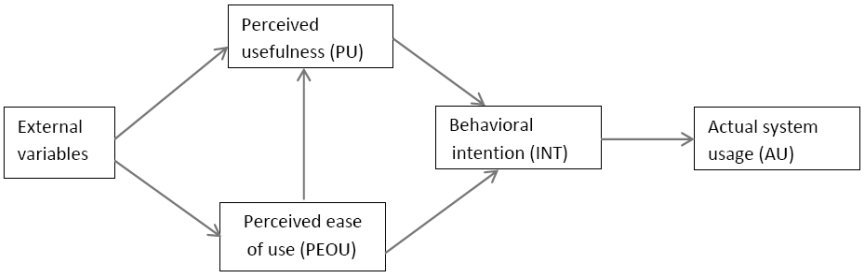
Source: Davis et al. (1989)

The initial model of TAM (Davis et al. 1989) (Figure 1) shows the relationships between perceived usefulness (PU), perceived ease of use (PEOU), attitude toward using the technology (ATT), behavioural intention to use (BI) and actual use (AU). Perceived usefulness (PU) refers to the degree to which the user believes that using the technology will improve his or her work performance in an organisational context. Perceived ease of use (PEOU) refers to the degree to which a user perceives the use of technology to be effortless. Perceived ease of use is also hypothesised to influence perceived usefulness. Both PU and PEOU are considered distinct factors influencing the attitude of the user towards using the technology. Previous studies show significant effect of perceived ease of use (PEOU) on perceived usefulness (PU) (Kleijnen et al., 2004; Wang et al., 2003). High level of PU influences attitude towards use strongly (Teo et al., 2008) and also influences the behavioural intention indirectly (Liu et al., 2005). Behavioural intention to use (BIU) determines whether users will actually use the system. BI is viewed as being jointly determined by a person's attitude towards using a system and perceived usefulness (Davis et al., 1989). The role of intention as a predictor of actual use is critical and has been well established information science and reference disciplines (Venkatesh et al., 2003). TAM presumes that behavioural intention is formed as a result of conscious decision-making process (Venkatesh et al., 2003). Empirical research shows direct and

significant relationship between behavioural intention and actual usage of technology (Mun and Hwang, 2003). Considerable evidence exists for stating that intention to perform behaviour predicts actual behaviour, and a majority of studies considering the technology adoption predict actual usage measuring only behavioural intention to use the system (Turner et al., 2010).

TAM has been modified over the years. Perceived ease of use and perceived usefulness were found to have a direct influence on behavioural intention and attitude towards using was eliminated. The final model of TAM includes external variables that influence the beliefs of a person towards a system and include system characteristics, user training, user participation in design and the nature of the implementation process (Venkatesh and Davis, 1996) (Figure 2).

Figure 2 Modified model of TAM



Source: Venkatesh and Davis (1996)

Studies report the usage of TAM across different contexts in various countries. A brief outline of extent literature is presented in the present discussion. Nick (2012) examines the perception of students towards podcasting as a review and preparatory tool for higher education using TAM. Results of the study indicated a positive relationship between perceived ease of use and perceived usefulness. A study investigating the use of e-books as learning material among undergraduates in Malaysia, showed perceived ease of use was positively related to perceived usefulness, while the perceived usefulness had a significant effect on attitude and intention to use e-books (Malathi and Rohani, 2011). Ahmed et al. (2011) examined the attitudes and intentions of students to use internet-based software using TAM and found that perceived usefulness, perceived ease of use predict attitude of students. Attitude was found to be a stronger predictor of the intention of students to use social software. Another study on the social media usage behaviour using TAM also shows a significant positive relationship between PEOU and PU (Rupak et al., 2014). PU and PEOU were found to have a positive effect on the behavioural intention; PEOU also resulted in higher behavioural intention; and attitude was found to influence the behavioural intention of students using websites for learning in Oman (Sujeet and Jyoti, 2013). While studying the e-book usage Bansal (2010) found perceived ease of use positively impacted perceived usefulness. A study on the adoption of social networks among students using TAM showed that PEOU and PU impact attitude and attitude in turn impacts behavioural intention to use (José Carlos and Ana Maria, 2011). Lee and Chang (2011) investigate the consumer attitude towards online co-design in mass customisation among Korean consumers using TAM. Their study indicated that PEOU positively affected PU while PU positively affected attitudes in the case on online mass customisation.

Similar to a study conducted by Al-Adwan et al. (2013) in Jordan, this study approaches e-learning as a system that uses internet technology to deliver information to students with interactions through computer interfaces. The current study analyses e-learning from the TAM perspective among Indian students and formulates the following hypotheses.

H1 Perceived ease of use of e-learning has a significant effect on the perceived usefulness of e-learning.

H2 Perceived ease of use of e-learning has a significant effect on attitude towards using e-learning.

H3 Perceived usefulness of e-learning has a significant effect on attitude towards using e-learning.

H4 Perceived usefulness of e-learning has a significant influence on the intention to use e-learning.

H5 Perceived ease of use of e-learning has a significant influence on intention to use e-learning.

H6 Attitude towards using e-learning has a significant influence on intention to use e-learning.

H7 Intention to use e-learning has a significant influence on actual use of e-learning.

H8 Perceived ease of use of e-learning has a significant effect on the actual use of e-learning.

H9: Perceived usefulness of e-learning has a significant effect on the actual use of e-learning.

3 Methodology

3.1 Data collection

The study is exploratory in nature and hence a convenience and judgmental sample was used. A structured and self-reported questionnaire was used for collecting data. Suggestions were taken from a group of three marketing professors and two information technology professors for eliminating any ambiguity in the questionnaire. The instrument used for the study was tested on a group of 15 students of Pune University in Nashik. Based on the responses of the experts and the test sample, minor changes in the language of the questions were made to arrive at the final questionnaire.

Ten graduate students were trained on the requirements of the study. These students assisted the researchers in collecting data. Researchers along with the team of graduate students personally administered the final questionnaires to the prospective respondents, pursuing post-graduation course in management, after briefing them about the study. The survey was conducted at various colleges offering post-graduation courses in Business Management, in the city of Nashik, Maharashtra, India. Questionnaires were handed over to prospective respondents only when he/she showed willingness to participate in the survey. The minimum required age of the respondents was 18 years, as they must be

students pursuing the post-graduation course and must have been using e-learning during their post-graduation course. Data was collected between November 2013 and January 2014.

3.2 Sample profile

Of the 120 students contacted only 116 respondents returned the filled questionnaires. Of these four questionnaires were discarded as they were not completely filled. The researchers were left with 112 usable questionnaires and the sample size for the study was 112 respondents (N = 112) (Table 1). Hence, the response rate for the study was 93%. The sample comprised of 62 male respondents (56% of respondents) and 50 female respondents (44% of respondents), showing no significant difference in the distribution of the respondents between the gender. Most students (67% of the respondents) were from the commerce background (chi-square 61.839). There was no significant difference in the respondents in terms of their monthly family incomes.

Table 1 Sample profile

Characteristics	*N*	*%*	*Comment*
Gender			
Male	62	56	Chi-square = 1.286
Female	50	44	p value = 0.257
Total	*112*	*100*	*Not significant*
Graduation in			
Commerce	75	67	Chi-square = 61.839
Computer science	28	25	p-value = 0.000
Engineering	9	8	
Total	*112*	*100*	*Significant*
Monthly family income			
Upto INR 25,000	26	23	Chi-square = 1.643
INR 25,001–INR 50,000	29	26	p-value = 0.650
INR 50,001–INR 100,000	33	29	
Above INR 100,000	24	21	
Total	*112*	*100*	*Not significant*

3.3 Construct description

The measurement items for TAM constructs (perceived usefulness, perceived ease of use, attitude and behavioural intention) as shown in Table 2 used for the purpose of this study comprised of 19 items. The instrument comprised of 15 items measured on a five-point Likert like scale, with the anchoring points ranging from 1 for strongly disagree to 5 for strongly agree. Two items were coded as 5 for very strongly disagree to 1 for very strongly agree. Item 16 measured the actual usage of e-learning by the respondent. Four items measuring perceived usefulness and four items measuring perceived ease of use were adapted from the scale developed by Davis et al. (1989). Items for measuring attitude towards e-learning and intention to use e-learning were adapted from Agarwal and Prasad (1999) and Venkatesh et al. (2003). Respondent scores for each of the factor

were arrived at by calculating the respective mean scores of items identified for each factor. Items numbered 17, 18 and 19 provided other demographic details of the respondent needed for the study such as – gender, stream of education during graduation and monthly family income.

4 Data analysis

4.1 Summary statistics

Summary statistics and frequency distributions were calculated for each of the 16 items of the construct as shown in Table 2. Means for the items range between 3.53 and 4.17 and standard deviations of items are less than 0.948.

Table 2 Summary statistics

Item description		*Mean*	*Std. deviation*
Perceived usefulness (PU)			
PU1	Using e-learning would enhance my effectiveness in learning.	3.84	.823
PU2	Using e-learning would improve my course performance.	3.86	.847
PU3	Using e-learning would improve my productivity in my course.	3.96	.752
PU4	I find e-learning useful for my studies.	3.87	.811
Perceived ease of use (PEOU)			
PEOU1	I find e-learning easy to use.	3.89	.764
PEOU2	Learning to use e-learning would be easy for me.	3.78	.846
PEOU3	My interaction with e-learning is clear and understandable.	3.59	.766
PEOU4	It would be easy for me to find the required information using e-learning.	3.87	.844
Attitude towards e-learning (ATT)			
ATT1	I dislike the idea of using e-learning. (R)	4.03	.799
ATT2	I have a generally favourable attitude toward using e-learning	3.73	.671
ATT3	I believe it is a good idea to use this e-learning for my course.	4.17	.848
ATT4	Using e-learning is a foolish idea.	3.88	.814
Behavioural intention to use (INT)			
INT1	I intend to use e-learning during the semester.	3.53	.859
INT2	I will return to e-learning often.	3.55	.804
INt3	I intent to visit e-learning frequently for my course.	3.64	.948
Actual use (AU)			
AU	I use e-learning frequently	3.87	.885

Note: Sample size: N = 112

4.2 Reliability statistics

Cronbach alpha which measures the internal consistency of the scale was calculated for each of the factor and for the entire scale (Cronbach alpha = 0.885) as shown in Table 3.

Internal consistency measures the inter-relatedness of the items used in the test. Hair et al. (2003) recommended Cronbach's alpha values from 0.6 to 0.7 as the limit of acceptability. A maximum alpha value of 0.09 has been recommended (Streiner, 2003). A very high value for alpha suggests that some items are redundant and they may test the same question. Cronbach alpha of 0.885 for the construct used in the current study indicates that it can be used for further analysis.

Table 3 Reliability statistics

Factor	*Cronbach's alpha*
Perceived ease of use (PEOU)	0.770
Perceived usefulness (PU)	0.710
Attitude towards e-learning (ATT)	0.615
Intention to use (INT)	0.605
Overall TAM scale (16 items)	*0.885*

4.3 Hypotheses testing

Nine hypotheses were formulated based on the initial TAM model developed by Davis et al. (1989). Each hypothesis was tested for significance based on the regression statistics. Table 4 presents the summary of the regression data for each hypothesis. Larger beta values were observed for larger t-values and smaller p-values across all hypotheses. A detailed discussion on each of the hypothesis follows.

Table 4 Regression statistics for the hypotheses

Hypothesis	*Independent variable*	*Dependent variable*	*Un-standardised coefficients*		*F*	*t*	*p*	R^2	*Hypothesis supported*
			β	*Std. error*					
H1	PEOU	PU	0.687	0.063	118.128	10.869	0.000	0.518	Yes
H2	PEOU	ATT	0.468	0.069	45.840	6.771	0.000	0.294	Yes
H3	PU	ATT	0.495	0.072	47.002	6.856	0.000	0.299	Yes
H4	PU	INT	0.723	0.079	83.225	9.123	0.000	0.431	Yes
H5	PEOU	INT	0.782	0.067	35.862	11.656	0.000	0.553	Yes
H6	ATT	INT	0.684	0.096	50.817	7.129	0.000	0.316	Yes
H7	INT	AU	0.395	0.124	10.177	3.190	0.002	0.085	Yes
H8	PEOU	AU	0.533	0.126	17.780	4.217	0.000	0.139	Yes
H9	PU	AU	0.378	0.138	7.500	2.739	0.007	0.064	Yes

4.3.1 Hypothesis 1

H1 Perceived ease of use of e-learning has a significant effect on the perceived usefulness of e-learning.

Regression analysis was conducted for testing the first hypothesis where perceived ease of use (PEOU) was an independent factor while perceived usefulness (PU) was a dependent factor. Regression analysis gave a p-value of 0.000, which indicates the

existence of a significant relationship between the factors PEOU and PU. The calculated value of R^2 for the regression equation was 0.518 indicating that the predictor factor perceived ease of use (PEOU) explains 51.8 % of perceived usefulness (PU), the dependent factor. This is an overall measure of the strength of association and does not reflect the extent to which any particular independent factor is associated with the dependent factor. The unstandardised β coefficient indicates the extent to which PEOU predicts PU (for every unit of increase in PEOU, a 0.687 unit increase in PU is predicted). The F statistic tests the full model against a model with no variables and with the estimate of the dependent variable being the mean of the values of the dependent variable. Value of the F statistic ranges between zero to an arbitrarily large number. The value of F statistic for the regression model was 118.128, indicating the significance of the regression model.

4.3.2 Hypothesis 2

H2 Perceived ease of use of e-learning has a significant effect on attitude towards using e-learning.

Results of the regression analysis for testing the second hypothesis with PEOU as an independent factor and attitude (ATT) as a dependent factor showed a significant relationship (p-value of 0.000). R^2 for the regression equation was 0.294 indicating the strength of association, in other words PEOU explains 29.4% of the factor attitude (ATT). The unstandardised β coefficient of 0.468 indicates the extent to which the variable PEOU predicts attitude towards e-learning (ATT). F statistic of 45.840 indicates the significance of the regression model.

4.3.3 Hypothesis 3

H3 Perceived usefulness of e-learning has a significant effect on attitude towards using e-learning.

Significant relationship ($p = 0.000$ and $t = 6.856$) was observed between the independent factor perceived usefulness (PU) and the dependent factor attitude towards using e-learning (ATT), using the regression equation. R^2 value of 0.299 indicates that the factor PU explains 29.9% of the factor ATT. The unstandardised regression coefficient which indicates the average change in the factor ATT for a unit change in the factor PU was 0.495. The regression model was significant as seen from the value of F-statistic (47.002).

4.3.4 Hypothesis 4

H4 Perceived usefulness of e-learning has a significant influence on intention to use e-learning.

Perceived usefulness (PU) was taken as an independent factor, while intention to use (INT) was taken as a dependent factor for testing this hypothesis using regression analysis. Results revealed a significant relationship between both the factors ($p = 0.000$ and $t = 9.123$). The factor PU explained 43.1% of the factor INT. Unstandardised coefficient 0.723 indicates the ability of the factor PU to predict the factor INT. F-statistic (35.863) for the regression model indicates that the model is significant.

4.3.5 Hypothesis 5

H5 Perceived ease of use of e-learning has a significant influence on intention to use e-learning.

Regression analysis was conducted to test this hypothesis, where the perceived ease of use (PEOU) was a predictor factor and intention to use e-learning (INT) was the dependent factor. Significant relationship was observed between the two factors ($p = 0.000$ and $t = 11.656$). The factor PEOU explained 31.6% of the factor INT. The regression model was observed to be significant ($F = 35.862$).

4.3.6 Hypothesis 6

H6 Attitude towards using e-learning has a significant influence on intention to use e-learning.

Attitude towards using e-learning (ATT) was found to have a significant influence on the intention to use e-learning (INT) ($p = 0.00$ and $t = 7.129$). The factor ATT explained 31.6% of the factor INT, while the unstandardised beta coefficient was 0.684 indicating the predictability of the factor INT. The regression model was significant ($F = 50.817$).

4.3.7 Hypothesis 7

H7 Intention to use e-learning has a significant influence on actual use of e-learning.

Regression analysis showed that intention to use e-learning was found to have a significant influence on the actual use of e-learning ($p < 0.005$ and $t = 3.190$). R^2 statistic of 0.085 indicates that the intention to use e-learning explains 8.5% of the actual use of e-learning. The unstandardised beta coefficient of 0.395 indicates that the predictability of the factor actual use (AU). The model reports significance ($F = 10.177$).

4.3.8 Hypothesis 8

H8 Perceived ease of use of e-learning has a significant effect on the actual use of e-learning.

Significant influence of perceived ease of use on the actual use was observed based on the regression analysis as $p < 0.001$ while the t-statistic was 4.217 indicating significance of the relationship. The factor perceived ease of use (PEOU) explained 13.9% of the factor actual use (AU). The F-statistic (17.780) indicates the significance of the model.

4.3.9 Hypothesis 9

H9 Perceived usefulness of e-learning has a significant effect on the actual use of e-learning.

A significant relationship was observed on the application of linear regression on the factor perceived usefulness as an independent factor and the actual use as a dependent factor ($p < 0.01$ and $t = 2.739$).

4.4 Mediating effect of various factors

Significant relationships were observed for all hypotheses. Hence the mediating effect of different factors was considered as shown in Table 5. Results comply with the mediation criteria of Baron and Kenny (1986). The effects of perceived ease of use on behavioural intention are mediated by perceived usefulness (Money and Turner, 2004). Consistent with past research, findings of this study show perceived usefulness mediates perceived ease of use and behavioural intention. While some studies (Chang and Cheung, 2001; Bruner and Kumar, 2005) state that attitude mediates the effect of perceived usefulness and perceived ease of use on behavioural intention, Venkatesh (1999) and Venkatesh and Davis (2000) did not find a full mediation effect of attitude. Attitude was found to mediate the belief-intention link in the current study. Intention completely mediates the effects of perceived usefulness, perceived ease of use on usage behaviour (Venkatesh and Davis, 2000). Complete mediation was not observed for intention on perceived usefulness, perceived ease of use on usage behaviour (AU) in this study. Partial mediation effect of intention on the relationship between attitude and behaviour was observed in a study on internet purchasing behaviour in Malaysia (Ilham and Nik Kamariah, 2012). A similar observation was reported in the current study.

Table 5 Mediating effect of various factors

S. no.	*Path A*	*Path B*	*Path C*	*Path D*	*Mediating factor*	*Mediation effect (complete/ partial)*	*Sobel p-value*	*Studies*
1	PEOU → INT	PEOU → PU	PU → INT	PEOU → PU → INT	PU	Complete	p = 0.0062 (p < 0.01)	Money and Turner (2004)
2	PEOU → INT	PEOU → ATT	ATT → INT	PEOU → ATT → INT	ATT	Complete	p = 0.0051 (p < 0.005)	Chang and Cheung (2001) and Bruner and Kumar (2005)
3	PU → INT	PU → ATT	ATT → INT	PU → ATT → INT	ATT	Complete	p = 0.0016 (p < 0.005)	
4	ATT → AU	ATT → INT	INT → AU	ATT → INT → AU	INT	Partial	p = 0.036 (p > 0.01)	Ilham and Nik Kamariah (2012)
5	PEOU → AU	PEOU → INT	INT → AU	PEOU → INT → AU	INT	Partial	p = 0.815 (p > 0.01)	Venkatesh and Davis (2000)
6	PU → AU	PU → INT	INT → AU	PU → INT → AU	INT	Partial	p = 0.073 (p > 0.01)	

5 Conclusions

This paper examines the applicability of TAM in the context of e-learning among the university students in India. It examines the relationships between the factors – perceived ease of use of e-learning (PEOU), perceived usefulness of e-learning (PU), attitude towards e-learning (ATT), behavioural intention to use e-learning (INT), and actual use (AU). These factors were adapted from the initial model developed by Davis et al (1989). Regression analyses were conducted to test for the applicability of TAM model in the e-learning context. Thus, the findings can be summarised as –

1 perceived ease of use has a significant effect on perceived usefulness, attitude, behavioural intention and actual use of e-learning

2 perceived usefulness has a significant effect on attitude, behavioural intention and actual use

3 attitude has a significant effect on intention to use

4 behavioural intention to use has a significant influence on actual use of e-learning.

Observations from the current study are in line with TAM (Table 4), which theorises that an individual's actual system usage is determined by behavioural intention, which is in turn jointly determined by perceived usefulness and perceived ease of use (Davis, 1989).

Results of the current study show similarities to studies by

1 Ronnie et al. (2011), who state that perceived ease of use had a significant influence on the perceived usefulness in their study on the e-portfolio system

2 Davis (1989) and Hu et al. (1999) who found perceived ease of use had a significant effect on attitude towards usage

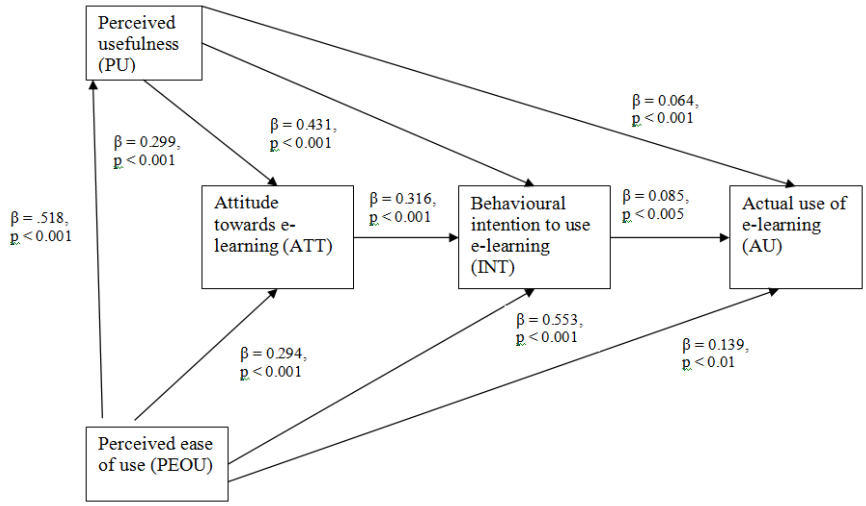
3 Teo (2009), who states that attitude is a significant predictor of the intention to use technology.

Davis (1989) states that perceived usefulness has a significantly greater correlation with usage behaviour than did perceived ease of use. However, the results of the current study showed perceived ease of use had a greater effect than perceived usefulness on the behavioural intention (R^2 values from Table 4).

Complete mediation was observed for perceived usefulness between perceived ease of use and intention to use. Perceived ease of use, perceived usefulness and intention were mediated by attitude. Similarly, behavioural intention mediated for attitude towards e-learning and actual use of e-learning. Figure 3 represents the summary of the model based on TAM for e-learning.

Findings of the study present important implications for educational institutions providing e-learning courses. Understanding what drives students in their adoption of e-learning technology is important for its success. Acceptance of e-learning by students would increase if the students perceive the use of e-learning to be simple and useful. This in turn would influence their attitude towards e-learning thereby influencing their intention to use e-learning more frequently.

Figure 3 Summary of the regression analysis based on TAM for e-learning



5.1 Limitations of the study

The current study uses self-reported use data. Researchers state that self-reported use data is subjective and unreliable in measuring actual use of a system (Yousafzai et al., 2007). The study focuses only on the students in one city. A larger sample with respondents representing different regions of the country can be used to generalise the results. Researchers argue that students may have different motivations such as obtaining grades, rewards and so on (Yousafzai et al., 2007; Legris et al., 2003; Lee et al., 2003 as referred by Chuttur, 2009). The study is subject to response bias and there was no treatment for non-responses. The study uses self-reported construct as the data needed for the study comprised of the perceptions of students for e-learning. Longitudinal studies can provide more generalised results.

5.2 Scope for further research

Legris et al. (2003) state that only 60% of studies on TAM considered the external variables. This study can further be extended to identify external variables influencing TAM and examine their impact on TAM in the e-learning context. An investigation into the computer self-efficacy of students would provide a greater understanding of their attitude towards e-learning. An investigation of the generalisability of computer self-efficacy to examine the TAMs was suggested by Thompson et al. (2006). TAM provides a systematic understanding of the intentions of the students to use e-learning. Such an understanding helps educators to examine their assumptions about the perceptions of the students for the acceptance of a new technology. Influence of the demographic variables on TAM in the Indian context is to be examined in further studies.

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